

Literature Status for arXiv:1504.04202: Ascoli Free Locally Convex Spaces

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Status

This is a `literature_already_answered` packet. It records that the free locally convex space question in the final section of Gabrielyan–Kakol–Plebanek, *The Ascoli property for function spaces and the weak topology of Banach and Frechet spaces*, arXiv:1504.04202, was answered by Gabrielyan, *On reflexivity and the Ascoli property for free locally convex spaces*, arXiv:1805.07028.

Source Question

The final section of arXiv:1504.04202, titled “Proof of Theorem `t:Ascoli-E-weak` and final questions” in the source file, asks the following two questions relevant to free locally convex spaces:

If $L(X)$ is Ascoli, is X Ascoli?

This is the $L(X)$ -case of the question labelled `q:Ascoli-Free` in the source TeX, which also asks the analogous questions for the free and free abelian topological groups $F(X)$ and $A(X)$.

The same final section then asks the sharper question labelled `q:Ascoli-Free-Discr`:

If $L(X)$ is Ascoli, is X countable and discrete?

The source adds that this was not known even under the stronger assumption that $L(X)$ is a $k_{\mathbb{R}}$ -space.

Supporting Answer

The abstract of arXiv:1805.07028 states that, answering a question posed in the GKP paper, it proves that $L(X)$ is Ascoli if and only if X is a countable discrete space. In the introduction the supporting paper repeats the exact question, citing GKP, and then states Theorem 1.2:

For a Tychonoff space X , $L(X)$ is Ascoli if and only if X is countable discrete.

This is precisely the affirmative answer to `q:Ascoli-Free-Discr`. Since every countable discrete space is Ascoli, it also gives an affirmative answer to the $L(X)$ -branch of `q:Ascoli-Free`. The supporting authors explicitly identify the result as answering the GKP question.

Scope

This packet records an already-known answer for the free locally convex space $L(X)$. It does not answer the $F(X)$ or $A(X)$ branches of **q:Ascoli-Free**, and it does not address the separate question **q:Ascoli-Free-F** asking for which metrizable spaces the groups $F(X)$ and $A(X)$ are Ascoli.

The theorem also implies the stronger $k_{\mathbb{R}}$ -variant mentioned after the source question in the following direction: if $L(X)$ is a $k_{\mathbb{R}}$ -space, then $L(X)$ is Ascoli, hence X is countable discrete by Theorem 1.2.

Search Evidence

The lane-1 deterministic queue selected arXiv:1504.04202. The cheap run indexes were searched for the arXiv id, title words, **Ascoli property**, **function spaces**, **free locally convex**, **L(X)**, and **Question 3.6**; no duplicate packet for this target was found. Local source searches for **L(X)**, **Ascoli**, **free locally convex**, **Ascoli**, and **question posed in GKP** identified arXiv:1805.07028 as the decisive later paper.

References

- [1] S. Gabrielyan, J. Kakol, and G. Plebanek, *The Ascoli property for function spaces and the weak topology of Banach and Frechet spaces*, arXiv:1504.04202.
- [2] S. Gabrielyan, *On reflexivity and the Ascoli property for free locally convex spaces*, arXiv:1805.07028.